

Lab 7: Introduction to Packet Tracer

Assignment Date: 19 July 2026 | Deadline: 27 July 2026
BSc.CSIT 4th Semester | New Summit College | Computer Network

Objective: To gain a basic understanding of Cisco Packet Tracer, its interface, and its essential functionalities.

1. Theory

1.1 Background Information

Cisco Packet Tracer is a network simulation and visualization tool developed by Cisco Systems. It allows users to design, build, configure and troubleshoot computer networks using virtual, drag-and-drop networking devices such as routers, switches, PCs and servers, without needing physical hardware. It is widely used for learning and practicing networking concepts such as IP addressing, routing, switching, VLANs and various topologies.

1.2 Key Concepts of Packet Tracer

- **Simulation-based learning:** Devices behave like real networking equipment, allowing configuration commands and cabling to be practiced safely.
- **Two operating modes:** Real-Time mode (network behaves instantly, like a real network) and Simulation mode (packets are shown moving step-by-step so their journey can be observed).
- **Device configuration:** Devices can be configured either through a GUI-based Config tab or through a CLI (command line interface) tab that mimics real Cisco IOS commands.
- **Logical and Physical workspace:** Networks can be built in a logical topology view or a physical view that represents real-world placement (e.g. inside buildings/racks).

2. Interface of Cisco Packet Tracer

- a **Workspace** — The large central canvas where devices are placed, connected with cables, and the topology is built and viewed.
- b **Toolbar** — Contains common tools such as select, move, delete, zoom, add note, and drawing tools used to edit the workspace.
- c **Device-Type Selection Panel** — Located at the bottom-left, it groups devices into categories (Routers, Switches, Hubs, End Devices, WAN Emulation, Connections, etc.).
- d **Device Configuration** — Clicking a device opens a window with Physical, Config and CLI tabs used to power on/off modules, set basic parameters, or type IOS commands.
- e **Real-Time and Simulation Mode** — Toggled from the bottom-right; Real-Time runs the network instantly while Simulation mode lets you send and step through individual packets.
- f **Network Component Icons and Labels** — Every device and connection is represented in the workspace with an icon and an editable label/name for easy identification.
- g **Options and Preferences** — Found in the top menu, allowing customization of appearance, grid, and simulation preferences.
- h **Activity Wizard** — A tool used by instructors to create guided, gradeable network activities for students (not typically used by students directly).

■ *[Insert Screenshot Here] — Packet Tracer home screen after installation*

■ *[Insert Screenshot Here] — labeled overview of the workspace, toolbar and device panel*

■ *[Insert Screenshot Here] — a device's configuration window (Physical / Config / CLI tabs)*

■ *[Insert Screenshot Here] — Real-Time vs Simulation mode toggle*

Conclusion

This lab gave a foundational understanding of Cisco Packet Tracer as a network simulation tool, including its workspace, toolbar, device panels and configuration options. This understanding of the interface forms the basis for building and testing more complex networks in the following labs.

Lab 8: Creating a LAN and Testing the Connectivity using Packet Tracer

Assignment Date: 19 July 2026 | Deadline: 27 July 2026

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Objective: To understand the basics of a LAN (Local Area Network) and implement one in Cisco Packet Tracer.

1. Theory

1.1 LAN & its Architecture

A LAN (Local Area Network) is a network that connects computers and devices within a limited geographical area, such as a home, office, or a single building. It is typically owned and managed by a single organization and allows devices to share resources such as files, printers and internet access. A basic LAN architecture consists of end devices (PCs, laptops) connected to a central intermediary device such as a switch or hub, which forwards data between the connected devices.

1.2 Components Used

- **End Devices (PCs):** The computers that send and receive data on the network.
- **Switch:** An intermediary device that connects multiple end devices and forwards data only to the intended recipient based on MAC addresses.
- **Copper Straight-Through Cable:** Used to connect end devices to the switch.

1.3 Network Diagram

■ *[Insert Screenshot Here] — simple network diagram showing 2 PCs connected to a switch*

2. Implementation Sequence

- Place two PC devices and one Switch in the workspace from the Device-Type Selection panel.
- Connect each PC to the switch using a copper straight-through cable, selecting the correct FastEthernet ports on both ends.
- Assign a static IP address and subnet mask to each PC through its Desktop → IP Configuration tab (e.g. PC0: 192.168.1.1/24, PC1: 192.168.1.2/24), keeping both on the same subnet.
- Wait until all connection link lights turn green, indicating the devices are properly connected and powered.
- Open the Command Prompt on one PC and run `ping <other PC's IP>` to test connectivity between the two nodes.

■ *[Insert Screenshot Here] — two PCs connected to a switch with link lights green*

■ *[Insert Screenshot Here] — IP configuration window of a PC*

■ *[Insert Screenshot Here] — successful ping reply between the two PCs*

Conclusion

By building a simple LAN with two PCs connected through a switch and successfully testing connectivity using the ping command, this lab reinforced the practical understanding of how devices within a local network are physically connected and logically addressed to communicate with each other.